Video Transcript - Part 10

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Hey everyone, welcome to part ten. In this video we're covering seven data preparation techniques that sit between raw data and a model that trains well: handling outliers, binning, transforming, encoding, one hot encoding, scaling, and shuffling. Each one is simple on its own, but knowing when to use it is what separates a messy pipeline from a clean one.

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Here's the plan. We'll take each technique in turn, starting with outliers, then binning, transforming, encoding and one hot encoding, scaling, and shuffling. At the end, we'll put them together into a typical preparation workflow.

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Let's start with outliers. An outlier is a data point that sits far away from the rest, like a ten million dollar house in a neighborhood of three hundred thousand dollar homes. Outliers can be genuine rare events, or mistakes like typos and sensor glitches. There are two common ways to spot them. The z-score method flags values more than about three standard deviations from the mean. The IQR method flags anything more than one and a half times the interquartile range below the first quartile or above the third quartile. Box plots show this visually.

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Once you've found outliers, you have four options. You can remove them, which makes sense when they're clearly errors. You can cap them, sometimes called winsorizing, by replacing extreme values with a limit like the ninety fifth percentile. You can transform the feature, which pulls extreme values closer in. Or you can keep them, because sometimes the outlier is exactly the signal you care about, like a fraudulent transaction. The key rule is to never delete outliers blindly. Always ask whether the value is an error or real information.

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Next is binning, also called bucketing. Binning groups continuous values into a small number of ranges. Instead of exact ages, you might use groups like under eighteen, eighteen to thirty five, thirty six to sixty, and over sixty. Equal width binning makes every bin cover the same range of values, while quantile binning makes sure each bin holds roughly the same number of data points. Binning can reduce noise, soften the effect of outliers, and help models capture non linear patterns. The trade off is that you lose some of the detail in the original values.

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Transforming means applying a mathematical function to change the shape of a feature. The most common reason is skewed data. If most values are small and a few are huge, like incomes or house prices, a log transform squeezes the large values down and spreads the small ones out, giving you something closer to a bell curve. Square root is a milder option, and Box-Cox is a flexible family of power transforms that finds a good setting for you. This helps models that expect roughly normal data, and it reduces the pull of outliers too.

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Encoding turns categories into numbers, because most algorithms can't work with text labels directly. The simplest approach is label encoding, also called ordinal encoding, which assigns an integer to each category. It works best when there's a real order, like small, medium, and large. If you use it on categories with no order, such as colors or cities, the model may wrongly assume that one is bigger than another, and that's where one hot encoding comes in.

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One hot encoding solves that problem. It creates a new column for each category, with a one if the row belongs to that category and a zero otherwise. So red, blue, and green become three columns, and each row has exactly one of them switched on. No order is implied, which makes it the right choice for things like colors, cities, or product types. The downside is that a feature with hundreds of categories creates hundreds of columns, which connects back to the curse of dimensionality we covered earlier.

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Scaling puts features on comparable ranges. Imagine salary in the thousands next to a feature between zero and one. Many algorithms would let the big numbers dominate. Normalization, or min max scaling, squeezes values into a range from zero to one. Standardization, or z-score scaling, shifts the data so it has a mean of zero and a standard deviation of one. Scaling matters most for distance based models like k nearest neighbors and support vector machines, and for anything trained with gradient descent, including neural networks. Tree based models usually don't need it. One tip: calculate the scaling values from the training data only, then apply them to the test data, so no information leaks.

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Finally, shuffling means randomly reordering your rows before training. Datasets are often sorted, maybe by date, or by class with all the positive examples together. If you split sorted data without shuffling, your training set might contain only one class and your validation set the other. A model that sees data in a fixed order can also learn the order itself instead of real patterns. For neural networks, it's common to reshuffle at the start of every epoch. The big exception is time series data, where order carries meaning, so don't shuffle it.

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Let's put it together. A typical workflow starts by handling outliers, then transforming skewed features, then binning where it makes sense, then encoding categories, then scaling numeric features, and finally shuffling before you train. This order isn't a strict rule, and you'll adjust it for your data and model, but it's a solid default.

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So to recap, outliers need a decision, not an automatic delete. Binning, transforming, and encoding reshape your features into forms a model can use, and scaling and shuffling keep training fair and stable. Get these steps right and your models will learn from your data far more effectively. Thanks for watching, and I'll see you in the next one.